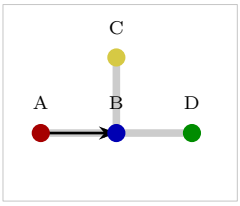
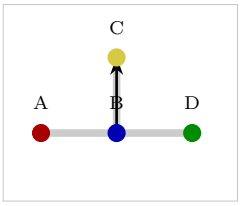
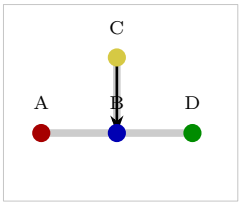
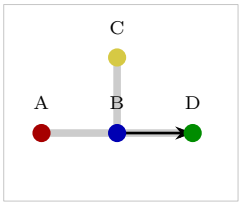
Move a pawn on the roads. Keep each journey as ordered step tiles; an adult may record. Remove only two adjacent steps along the same road in opposite directions. Replay the other steps in order. The map, start, and finish stay fixed.

input



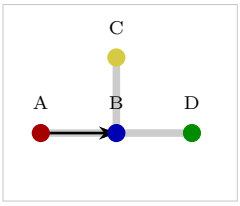


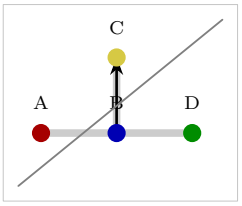


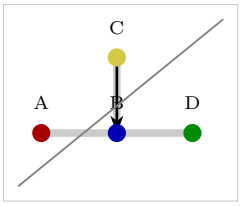


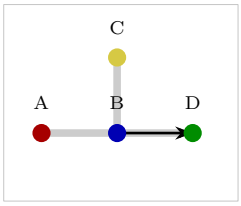
replay

remove



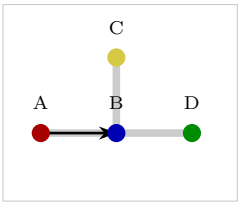


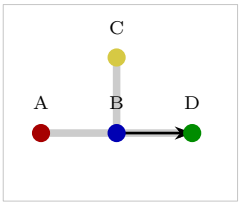




then replay

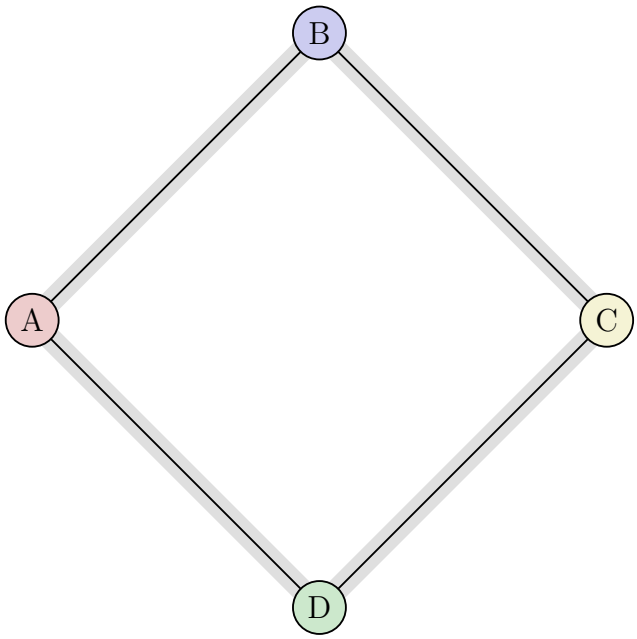
output





No steps left means the pawn stays at its start.

Problem 4: Shorten each recorded trip in different orders. Can the choice of first reversal ever change the final route on any road map? Explain, or give a counterexample.

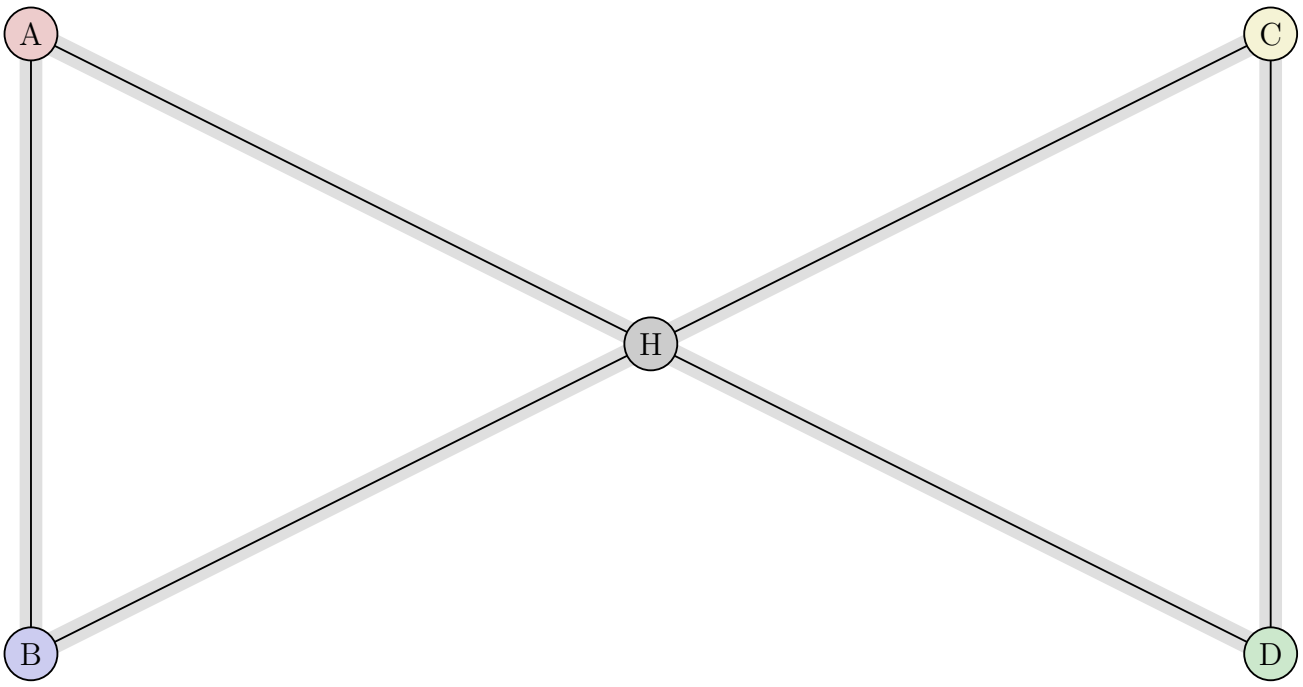


$A \to B \to A \to B \to C \to B \to A \to D \to C \to D \to A \quad \longrightarrow$ _____

$A \to B \to A \to D \to C \to B \to C \quad \longrightarrow$ _____

$A \to B \to C \to D \to A \to B \to C \to B \to A \quad \longrightarrow$ _____

Problem 5: From H, go around the left ring through A, then the right ring through C, then the left ring backward, then the right ring backward. Can the whole trip disappear by removing immediate reversals?



Problem 6: This trip goes around each ring once each way. Does that information alone tell whether a trip can disappear? Make another trip with those same counts and a different result.